AQUACHECK

Portable Multi-Parameter Water Quality Monitoring & Assessment Device

Project Title: AquaCheck Handheld Water Analyzer	**Target Platform:** ESP32 Dual-Core (Xtensa 240MHz)
Academic Scope: Embedded Systems Mini-Project	**Target Domain:** Environmental & Public Health Sensing
Estimated Budget: LKR 14,480 (~$47.50 USD)	**Output Interfaces:** 0.96" OLED, MicroSD Logger, Piezo Alert

1. Executive Summary & Problem Statement

Access to verified clean drinking water is a critical humanitarian priority across developing nations and specifically in Sri Lanka's dry zones, where agrochemical runoff and elevated fluoride/mineral concentrations are strongly linked to chronic health crises such as CKDu (Chronic Kidney Disease of unknown etiology). Conventional water testing depends on centralized laboratories. While accurate, laboratory analysis incurs 24–72 hour delays, prohibitive operational costs, and complex sample handling, preventing rapid, on-site diagnostics by field health officers, rural community well owners, and agriculturalists.

AquaCheck resolves this bottleneck by integrating a complete analytical instrument into an ultra-portable, battery-operated handheld device. It synchronously measures four primary physicochemical water quality parameters: **pH** (acidity/alkalinity), **Turbidity** (optical clarity/suspended solids), **Total Dissolved Solids (TDS/EC)** (dissolved ionic species), and **Water Temperature**. Powered by an ESP32 microcontroller, AquaCheck executes real-time temperature compensation, automatically classifies water safety against WHO and Sri Lanka Standards (SLS 614:2013), writes continuous logs to an onboard MicroSD card, and triggers immediate audio-visual danger alerts.

2. Project Objectives & Engineering Novelty

- **Synchronous Multi-Sensor Co-Location:** Acquire stable readings from 4 distinct physical probes submerged in a common liquid sample within 5 seconds.
- **Real-Time Embedded Compensation:** Execute onboard mathematical temperature compensation for both pH (Nernst equation) and electrical conductivity (EC/TDS at 25°C) to eliminate thermal drift.
- **Full Field Autonomy:** Deliver 8+ hours continuous runtime via an internal 18650 Li-ion cell with integrated USB-C charging and step-up power management.
- **Edge Health Classification:** Automatically benchmark sample measurements against SLS 614:2013 standards, outputting immediate 'Safe', 'Permissible', or 'Hazardous' status.

Engineering Novelty: Unlike generic IoT projects that push unverified analog readings to the cloud, AquaCheck resolves critical hardware challenges: (1) *Inter-probe electrical interference* where active TDS AC pulses corrupt high-impedance pH glass probes; (2) *Non-linear optical dispersion* in low-cost turbidity sensors; and (3) *Autonomous offline logging* with zero cloud dependency.

3. Technical Hardware Architecture & Interfacing Table

Subsystem / Module	Component Model	ESP32 Pin	Signal Type / Protocol	Operating Voltage
pH Measurement	Analog pH V1.1 (BNC Probe)	GPIO 32 (ADC1_CH4)	Analog (0.0V – 3.3V mapped)	5.0V / 3.3V Logic
Turbidity / Clarity	TS-300B Optical Turbidity	GPIO 33 (ADC1_CH5)	Analog (0.5V – 4.0V mapped)	5.0V VCC
TDS / Conductivity	Gravity Analog TDS Sensor	GPIO 34 (ADC1_CH6)	Analog AC Excitation (0–2.3V)	3.3V – 5.0V
Water Temperature	DS18B20 Stainless Probe	GPIO 4 (4.7kΩ Pull-up)	Maxim 1-Wire Digital Bus	3.3V VCC
Display Output	0.96" I2C OLED (SSD1306)	GPIO 21 (SDA), 22 (SCL)	I2C Bus (400 kHz)	3.3V VCC
Offline Storage	MicroSD SPI Breakout	GPIO 18, 19, 23, 5 (CS)	SPI Serial Bus (10 MHz)	3.3V VCC
Power Management	TP4056 + MT3608 Boost	VIN / 5V Rail	Regulated 5.0V & 3.3V Rails	3.7V Li-ion to 5.0V
Audio-Visual Alerts	Active Piezo + Bi-Color LED	GPIO 25 (Buzzer), 26, 27	Digital Output / PWM	3.3V Logic

4. Physical Models, Calibration & Compensation Algorithms

To achieve laboratory-grade reliability with low-cost analog transducers, AquaCheck integrates deterministic chemical and optical physics compensation routines into its embedded firmware loop:

A. Temperature-Compensated Electrical Conductivity (EC) and TDS:

The mobility of dissolved ions in water increases with temperature at approximately 1.9% per °C. Standard TDS readings must be normalized to a standard 25.0°C baseline reference. The compensation formula is:

$$EC_{25} = EC_{measured} / [\ 1.0 + 0.019 \times (T_{water} - 25.0)\ ]$$

Total Dissolved Solids (TDS in parts-per-million) is subsequently computed using the conversion factor:

$$TDS\ (ppm) = EC_{25}\ (\mu S/cm) \times 0.50$$

B. Nernstian Temperature Slope Scaling for pH:

The theoretical output voltage of a glass pH electrode is governed by the Nernst Equation. The millivolts generated per pH unit (the electrode sensitivity slope S) is directly proportional to absolute thermodynamic temperature:

$$S(T) = (2.3026 \times R \times T_{Kelvin}) / F$$ [= 54.20 mV/pH at 0°C; 59.16 mV/pH at 25°C; 62.13 mV/pH at 40°C]

AquaCheck applies a two-point calibration using standard buffer solutions (pH 4.01 and pH 6.86):

$$pH_{corrected} = 7.0 + [\ (V_{neutral} - V_{probe}) / S(T)\ ]$$

C. Non-Linear Polynomial Characteristic for Optical Turbidity:

The optical turbidity sensor utilizes an infrared transmitter-phototransistor pair. Light scattering caused by suspended particulate matter decreases the received phototransistor collector voltage non-linearly:

$$Turbidity\ (NTU) = -1120.4 \times (V_{sensor})^2 + 5742.3 \times V_{sensor} - 4352.9$$ (for $V_{sensor} < 2.50V$)

(When $V_{sensor} \geq 2.50V$, Turbidity is mapped to < 0.5 NTU denoting crystal-clear water).

5. Water Classification Matrix (SLS 614 & WHO Standards)

Water Quality Parameter	Ideal / Excellent	Permissible Limit (SLS 614)	Hazardous / Unsafe	Sensor Resolution
pH (Acidity / Alkalinity)	6.5 – 7.5	6.5 – 8.5	< 6.0 or > 9.0	±0.1 pH units
TDS (Dissolved Solids)	< 300 ppm	300 – 500 ppm	> 1000 ppm (Saline/Hard)	±10 ppm
Turbidity (Water Clarity)	< 1.0 NTU	1.0 – 5.0 NTU	> 5.0 NTU (Muddy/Suspended)	±0.5 NTU
Temperature	20°C – 28°C	15°C – 35°C	> 40°C (Abnormal thermal)	±0.25°C

6. Firmware Execution Flow & FreeRTOS Task Structure

To ensure deterministic sensor sampling without UI stutter, the ESP32 firmware is divided into two concurrent FreeRTOS tasks:

• **Task_SensorAcquisition (Core 0):** Sequential time-multiplexed ADC reading (pH, then Turbidity, then TDS), DS18B20 1-Wire interrogation, temperature compensation math, and alert threshold checking.

• **Task_UI_and_Logging (Core 1):** SSD1306 128x64 graphical rendering at 2 Hz and CSV data serialization to MicroSD card.

```
void Task_SensorAcquisition(void *pvParameters) {
for(;;) {
digitalWrite(TDS_POWER_GATE, LOW); // 1. Isolate TDS AC signal to eliminate noise on pH probe
float tempC = readDS18B20(); // 2. Sample liquid temperature baseline
float rawPH = readCalibratedPH(tempC); // 3. Sample pH with dynamic Nernst compensation
float turbidity = readTurbidityNTU(); // 4. Sample optical clarity
digitalWrite(TDS_POWER_GATE, HIGH); // 5. Enable TDS excitation only during its dedicated window
vTaskDelay(pdMS_TO_TICKS(50)); // Stabilize AC excitation
float tdsPPM = readCompensatedTDS(tempC); // 6. Calculate 25°C-normalized TDS
digitalWrite(TDS_POWER_GATE, LOW); // 7. Re-isolate TDS probe
evaluateSafetyLimits(rawPH, tdsPPM, turbidity); // 8. Check SLS 614 thresholds
vTaskDelay(pdMS_TO_TICKS(1000)); // Sample interval: 1.0 second
}
}
```

7. Comprehensive Bill of Materials (BOM) & Budget (Sri Lanka Market)

Current market prices collected from verified Sri Lankan electronic suppliers (Tronic.lk, Duino.lk, Microchip.lk):

Component / Module	Qty	Est. Unit (LKR)	Total (LKR)	Recommended Local Supplier
Analog pH Sensor Kit V1.1 (Probe + Signal Board)	1	4,200	4,200	Tronic.lk / Duino.lk
Analog Optical Turbidity Sensor Module (TS-300B)	1	2,850	2,850	Duino.lk / Microchip.lk
Gravity Analog TDS Sensor Module + Probe	1	1,950	1,950	Tronic.lk / Duino.lk

DS18B20 Waterproof Stainless Steel Probe	1	550	550	Microchip.lk / Tronic.lk
ESP32 DevKit V1 30-Pin Microcontroller Board	1	1,100	1,100	Duino.lk / StemLabs
0.96" I2C OLED Display (SSD1306 128x64)	1	850	850	Local Hobby Stores / Daraz
MicroSD Card SPI Module + 8GB Class 10 Card	1	900	900	Daraz.lk / Local Shop
18650 Li-ion Cell (2600mAh) + TP4056 USB-C Board	1	750	750	Tronic.lk / Duino.lk
MT3608 DC-DC Step-Up Boost Module (3.7V to 5V)	1	180	180	Duino.lk / Tronic.lk
Power Gating MOSFET (2N7000) & Passives	Lot	350	350	Pettah Electronics Market
Handheld ABS/Acrylic Enclosure & Probe Wand	1	800	800	Makerspace 3D Print / Acrylic
TOTAL ESTIMATED PROJECT BUDGET:	--	--	**LKR 14,480**	**~$47.50 USD (Sub-$50 Target)**

8. Project Implementation Roadmap & Milestones (6-Week Gantt Structure)

Phase & Timeline	Key Engineering Milestones & Deliverables	Measurable Verification Criteria
Week 1: Procurement & Subsystem Testing	Procure sensors and components from local vendors. Test individual analog outputs with breadboard sketches.	Clean oscilloscope/serial ADC stream from each sensor independently.
Week 2: Calibration & Compensation Lab	Perform 2-point pH buffer calibration (pH 4.01/6.86). Calibrate TDS sensor using standard NaCl calibration solutions.	Measurement error < 2.5% compared to known chemical reference standards.
Week 3: Dual-Core FreeRTOS Architecture	Develop FreeRTOS tasks: Sensor acquisition, Nernst/EC math routines, OLED screen layout, and CSV data logging.	2 Hz stable display refresh rate with zero dropped sensor readings.
Week 4: Power & Hardware Integration	Integrate TP4056 USB-C charging, 18650 cell, and MT3608 boost converter on custom perfboard / prototype PCB.	Sustained battery runtime > 8 hours under active continuous sampling.
Week 5: Field Sample Validation	Collect and test 5 diverse water samples: Distilled water, municipal tap water, agrochemical well water, pond, and saline solution.	Statistically distinct classification of safe vs contaminated samples.
Week 6: Enclosure Assembly & Final Defense	Mount electronics into handheld casing with probe wand holder. Finalize project proposal report and demonstration rig.	Functional, self-contained prototype ready for live evaluator demonstration.

9. Live Demonstration & Verification Testing Protocol

A compelling project defense hinges on demonstrating clear differentiation between real liquid samples. The following 4-vial evaluation protocol will be executed during the project examination:

Sample Test Vial	Expected pH	Expected TDS	Expected Turbidity	System Classification Verdict
Vial A: Deionized / Bottled Water	6.8 – 7.2	< 50 ppm	< 0.5 NTU	SAFE (Green LED, Clean)
Vial B: Municipal Tap Water	7.0 – 7.5	150 – 250 ppm	< 2.0 NTU	SAFE / PERMISSIBLE
Vial C: Muddy Runoff / Silt Water	6.5 – 7.8	200 – 400 ppm	> 25.0 NTU (Cloudy)	UNSAFE (Turbidity Alert)
Vial D: High-Saline / Mineral Well	7.5 – 8.8	> 1200 ppm (High)	< 3.0 NTU	HAZARDOUS (Mineral Overload)

10. Engineering Challenges, Failure Modes & Mitigations

• **Challenge 1: Cross-Probe Electrical Noise (Galvanic Coupling):** The TDS sensor applies an active AC excitation waveform into the fluid. When the high-impedance glass bulb pH probe is placed in the same liquid, this excitation introduces severe millivolt offsets, corrupting the pH reading.

Mitigation Strategy: Hardware Power Gating. The ESP32 drives an N-channel MOSFET switch (2N7000) that completely cuts power to the TDS excitation circuitry while the pH probe is being sampled. The sensors are never powered concurrently.

• **Challenge 2: ESP32 ADC Non-Linearity at Signal Margins:** The built-in 12-bit SAR ADC on the ESP32 exhibits significant non-linear saturation below 0.12V and above 2.80V.

Mitigation Strategy: Signal Conditioning & Multi-Point Calibration. All sensor amplifier boards are biased so their operational range falls between 0.40V and 2.40V (the linear region). Furthermore, the ESP-IDF `esp_adc_cal` library with eFuse Vref calibration is utilized.

• **Challenge 3: Sensor Bio-Fouling & Probe Degradation:** Submerging optical and glass sensors continuously leads to algal film deposition and electrolyte depletion in the pH reference junction.

Mitigation Strategy: Dip-and-Rinse Operational Protocol. AquaCheck is engineered as an on-demand spot tester rather than a continuous immersion buoy. A protective cap with 3M KCl soaking sponge is provided for probe storage.

11. Expected Deliverables & Academic Sign-off

Upon successful project completion, the following deliverables will be formally submitted for evaluation:

1. Fully integrated, handheld AquaCheck prototype housed in a durable 3D-printed enclosure.
2. Complete open-source firmware repository containing calibration math, FreeRTOS tasks, and OLED UI code.
3. Experimental validation report analyzing accuracy benchmarks against certified reference buffer solutions.
4. Project Demonstration Video and Technical Documentation Dossier.

Prepared By:	**Endorsed By:**	**Department Approval:**
_____	_____	_____
Student Investigator / Candidate	Project Supervisor / Advisor	Head of Department / Examiner
Date: ____ / ____ / 2026	Date: ____ / ____ / 2026	Date: ____ / ____ / 2026